

IO391

Simulink Driver Block Manual

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1 Description

1.1 IO391

Introduction

Configurable FPGA I/O modules provide a flexible and cost-effective way to configure and reconfigure I/O and protocol functionalities for a set of FPGA code modules delivered as part of a custom implementation package.

Custom implementation packages, once purchased from Speedgoat, can be downloaded from the Speedgoat Customer Portal and include the following items:

- **Documentation:** Comprehensive documentation including installation steps, descriptions of implemented functionality, pin mapping diagrams and explanations on how to get started with demo model examples
- **Simulink® Library:** A Simulink library, including links to the main Speedgoat Library, and which clearly shows the version of the driver blocks that match with your specific FPGA bitstream
- **Simulink demo model:** A simple demo model illustrating how the implemented code module functionality is used, with loopback wirings wherever possible
- **FPGA bitstream:** A *.mat file including the FPGA bitstream content used to configure the FPGA as well as additional meta data used by the Simulink driver blocks

Driver Blocks

- IO391-50k - Setup
- FPGA code modules

Pin Mapping

- Pin mapping for IO391

2 Pin Mapping

2.1 IO391 Pin Mapping

Terminal Board A: digital I/O

Pin	Signal	Transceiver
1a	Ground	
2a	Ground	
3a	FPGA code module functionality	TTL
4a	FPGA code module functionality	TTL
5a	FPGA code module functionality	TTL
6a	FPGA code module functionality	TTL
7a	FPGA code module functionality	TTL
8a	FPGA code module functionality	TTL
9a	FPGA code module functionality	TTL
10a	FPGA code module functionality	TTL
11a	FPGA code module functionality	TTL
12a	FPGA code module functionality	TTL
13a	FPGA code module functionality	TTL
14a	FPGA code module functionality	TTL
15a	0V	
16a	5V	
17a	Ground	

Note

Note: 0V is internally connected to the ground of the target machine whereas analog ground is connected to the ground of the I/O module. 5V are supplied by the real-time target machine, which is capable of providing 1A continuous current. For more information, refer to the real-time target machine User Manual.

Terminal Board B: digital I/O

Pin	Signal	Transceiver
1b	Ground	
2b	Ground	
3b	FPGA code module functionality	TTL
4b	FPGA code module functionality	TTL
5b	FPGA code module functionality	TTL
6b	FPGA code module functionality	TTL
7b	FPGA code module functionality	TTL
8b	FPGA code module functionality	TTL
9b	FPGA code module functionality	TTL
10b	FPGA code module functionality	TTL
11b	FPGA code module functionality	TTL
12b	FPGA code module functionality	TTL
13b	FPGA code module functionality	TTL
14b	FPGA code module functionality	TTL
15b	FPGA code module functionality	TTL
16b	FPGA code module functionality	TTL
17b	Ground	

3 Driver Blocks

3.1 IO391-50k - Setup

Library

Library: Speedgoat



Description

The Setup block dialog box can be used to configure all the supported I/O types of this module, and therefore impacts on all the other driver blocks.

Your model must contain one Setup block for each configurable FPGA I/O module in your target machine.

Ports

This driver block has no input or output ports.

Parameters

Module setup Tab

IO3XX Module

Select the name of the FPGA-based I/O module for which you are configuring the setup parameters.

IO3XX Module ID

Given that a target machine can contain more than one configurable FPGA I/O module, this parameter is used by the other driver blocks to identify a specific module. Select a unique number (1-10) for each of your modules. If your target machine only contains one FPGA module, leave the default value (1).

FPGA Bitstream

This parameter defines the FPGA bitstream to download to the FPGA of your configurable FPGA I/O module. A bitstream is a configuration file for the FPGA and defines the pin mapping and the embedded code modules. Refer to the custom implementation manual for the name of your custom bitstream and the custom I/O pin mapping. The FPGA bitstream is downloaded to the FPGA when your Simulink model is built and sent to the target machine.

Revision

This field shows the bitstream revisions. The field is only visible when there multiple revisions of the selected bitstream.

Timestamp

This field shows the timestamp information of the selected bitstream.

FPGA Frequency/Tick Time

This field shows the FPGA base clock and the corresponding time of one FPGA tick for the selected bitstream. These values should be used to calculate time-related parameters in the FPGA code modules.

PCI Slot (-1: autosearch)

If only one module of this type is present in the target machine, leave the default setting ("-1") to automatically locate the board. If more than one FPGA I/O module is present, enter the bus number and the PCI slot number of the board linked to this driver block using the "[BusNumber, SlotNumber]" format. To determine the PCI bus and slot numbers, type "speedgoat_io" in the MATLAB command window.

IO39X-50k Pull Resistor Tab

This tab is enabled if the Rear I/O conditioning module parameter is set to IO391-50k, IO393-50k or IO397-50k.

I/O Lines Port 1 Tab**Pull Resistor**

Select the pull resistor of the digital TTL I/O lines of port 1.

I/O Voltage

When the pull resistor parameter is not set to pull up, select the I/O voltage of the 8 digital I/O lines of port 1.